

**SCIENTIFIC REPORT
TWELFTH INDIAN EXPEDITION
TO ANTARCTICA**

TECHNICAL PUBLICATION NO. 10



सत्यमेव जयते

**DEPARTMENT OF OCEAN DEVELOPMENT
CGO COMPLEX, LODHI ROAD
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INDIA**

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FOREWORD

The Twelfth Scientific Expedition to Antarctica added new dimensions to the Indian Antarctic Programme. Continuation of the scientific work in Antarctica has led to further refinement in our understanding of some of the salient features of this continent and the surrounding ocean. The country has developed expertise in the diverse disciplines of polar sciences such as biology, earth, ocean, atmosphere, environment, medical and cold region engineering.

The 56 member team with 25 members winter component drawn from various research institutions/organisations and services were given time targeted scientific and logistic tasks. For the first time, this expedition was able to drill and retrieve a 60 m long ice core from Central Dronning Maud Land for conducting studies in paleo-environment and paleo-climate. Thus, a new dimension to Indian polar research was added. This highly motivated and dedicated team could successfully complete all routine tasks and established several new infrastructural facilities like construction of genset accommodation and commissioning of the new power house. This will further serve to strengthen the logistic support required for conducting front ranking scientific experiments from Maitri.

The present report is a compilation made by Dr. V.K. Dhargalkar, Leader of the Expedition of findings and experiences of the expedition members. I am sure and confident that this technical report will prove to be a valuable contribution in the field of Polar Sciences.

Dr. A.E. MUTHUNAYAGAM

Secretary, Govt. of India

Dept. of Ocean Development

New Delhi

January, 1996

PREFACE

The Twelfth Indian Scientific Expedition to Antarctica comprised 56 members drawn from 11 scientific institutions/organisations and three wings of defence services was set to achieve yet another goal of our National Antarctic Programme. The main tasks of the expedition were ice core drilling, construction of genset accommodation, commissioning of the new power house and continuation of efforts for further advancement of the multi-disciplinary scientific fields. The highly motivated and self-disciplined Twelfth Expedition team is a fine and unique example of coordination, cohesion and a team work of high standard. The team members irrespective of their status and affiliation joined hands to achieve against all odds the defined objectives. This speaks for dedication, mutual understanding, sense of duty and responsibility in execution of tasks in a most befitting manner in an unassuming continent of ice and snow.

Since the time of sailing from Goa, we have been receiving messages and greetings from well wishers, families and friends. This expedition has received very encouraging message from Shri P.R. Kumarmangalam, Honourable Minister of State for Science and Technology and Ocean Development. We thank him for a very thoughtful message and his interest in the Antarctic Science. Shri Ravi Naik, Chief Minister of Goa, Shri Simon D'Souza, Dy. Speaker, Goa Assembly, Dr. B.N. Desai, Director, National Institute of Oceanography, Goa, Dr. P. Rama Rao, Secretary, Shri J.V.R. Prasada Rao, Joint Secretary and other officials of Department of Ocean Development, families and friends who were present to bid us bon-voyage at Mormugao Harbour with blessings and good wishes, to them we owe a lot.

Messages of greetings and encouragements were received during our winter stay at Maitri and particularly during mid-winter day, when one looks for encouragement and good wishes from near and dear ones. Messages were received from Shri P.V. Narasimha Rao, Honourable Prime Minister of India. Shri Bill Clinton, President of United States of America, Minister of State for Science & Technology, Heads of all the three defence services, all the Heads of the participating institutions and families and friends have given us great encouragement, a pleasant feeling of being remembered and a source of inspiration to face the challenges in accomplishing our scientific goals.

Help rendered by Capt. A. Lennart Hedlund, Master of *M.V. Thuleland* and his officers and crew for efficiently assisting us in cargo handling, helicopter operation and in marine science programme is greatly acknowledged. We

record our sincere appreciation to Capt. Ruslan A. Zaynigabdinov, Master and crew members of *M.V. Stepan Krashninnikov* for extending all help in completing scientific and logistic tasks. In spite of the severe damage to the ship's rudder, they have displayed great skill, courage, seamanship and patience in bringing us safely to Durban Port in South Africa.

Our thanks are also due to Mr. Valeri Fedorov, Leader and members of the Novolazarevskaya station for cooperation and helping us in all possible ways. We are thankful to our neighbouring stations in Antarctica, *i.e.*, Syowa, Japan; SANAE, South Africa; George Von Neumayer, Germany for sending us greetings and exchanging scientific information particularly during the ROSA-93, ozone hole phenomenon programme.

For the first time in the history of Indian Scientific Expedition to Antarctica, XII Expedition has been planned with about 50% strength and short summer period of 49 days. This kept every one of us on toes to complete the summer programme within a stipulated time. The cohesive, hard working and sincere group of XII Expedition could accomplish all the summer tasks successfully. Army team accomplished herculean task of completing genset accommodation and commissioning of new power house 'Aditya' in time. The other logistic support in maintenance of station systems, vehicles, power house, support to scientific work and medical cover throughout the stay enabled this expedition to complete all the scientific and logistic tasks successfully. Indian Navy, in charge of communication, commissioned new INMARSAT terminal at Maitri and provided uninterrupted communication to the expedition. Indian Air Force showed great skill in flying out men and material particularly the genset accommodation in record time. Without which we could not have commissioned the new power house 'Aditya' in time. I would like to put on record my appreciation and thanks to all the three forces of defence for their dedicated services during the expedition.

All my scientific colleagues who have rendered helping hand in loading/unloading in their spare time, to them I record my appreciation. Dr. B.S. Ingole needs special appreciation for holding the fort in my absence from the ship and organising the flying schedules and cargo most effectively. He rose to the occasion in keeping up loading/unloading schedules most systematically and efficiently.

I record my gratefulness to several experts in different disciplines of science for reviewing the technical papers of this report and offering suggestions and modifications to improve quality and value of the papers. Thanks are also due to Dr. A.H. Parulekar, Dy. Director, NIO, Goa for the editorial scrutiny of the manuscript and to all my colleagues in NIO who have directly or indirectly helped me in bringing out this report in the present form. Thanks are

also due to the leaders and members of the IX and XI wintering team for cooperation during handing and taking over of the Maitri Station.

I am thankful to Dr. A.E. Muthunayagam, Secretary, Shri J.V.R. Prasada Rao, Joint Secretary and Dr. A. Mitra, SSO, Department of Ocean Development, New Delhi for their encouragement and interest to bring out this publication.

Last but not least, my sincere thanks to all the family members of the X winter over team for patience, faith and fortitude shown in tolerating our long absence and handling our affairs back home. In a real sense this is more of an expedition for our family members who are left behind to look after our near and dear ones. I may fail in my duty if I do not put on record my sincere appreciation and gratitude to all the members of X winter over team for performing beyond their capabilities by sheer zeal, devotion and dedication to work. I am confident that the commendable performance of this team will be lauded and will be remembered for years to come in the annals of Indian Antarctic history. I must consider myself lucky and fortunate to have been associated with such wonderful group of people and to have led them.

MAY GOD BLESS THEM ALL!

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VINOD K. DHARGALKAR

CONTENTS

	<i>Page</i>
Foreword	v
Preface	vii
The Twelfth Indian Scientific Expedition to Antarctica — Events & Achievements	1
<i>V.K. Dhargalkar</i>	

ATMOSPHERIC SCIENCES

Meteorological Studies Carried out by IMD Team during XII Antarctic Scientific Expedition at Maitri	23
<i>S. Venkateswarlu and R.P. Lal</i>	
HF Packet Beacon Experiment from Antarctica	41
<i>H.K. Gaur, V. Dwivedi and D.K. Gangopadhyay</i>	
Response of Geomagnetic Field at Indian Antarctic Station, Maitri to Increasing Electromagnetic Disturbance in Geospace . . .	47
<i>A.N. Hanchinal, K. Jeeva, Ajay Dhar and Girija Rajaram</i>	
Study of Seasonal Variations in the Quiet Time Geomagnetic Field at the Indian Antarctic Station Maitri	59
<i>A.N. Hanchinal, K. Jeeva, Ajay Dhar and Girija Rajaram</i>	

EARTH SCIENCES

An Attempt at Shallow Ice-core Drilling in Polar Continental Ice, Central Dronning Maud Land, Antarctica	77
<i>A. Chaturvedi and R. Asthana</i>	
Monitoring of Icebergs in Antarctic Waters	101
<i>M.P. Gaur, R. Asthana and A. Chaturvedi</i>	
Notes on Pattern of Snow Accumulation/Ablation on Iceshelf and Secular Movement of Dakshin Gangotri Glacier Snout in Central Dronning Maud Land, East Antarctica	111
<i>R. Asthana, M.P. Gaur and A. Chaturvedi</i>	

	<i>Page</i>
A Study of Thermal Behaviour of Different Ground Media in Response to the Polar Cold Front in Antarctica	123
<i>A. Chaturvedi and R. Asthana</i>	
Textural Studies of Retrograded Ductile Shear Zones from Schirmacher Hills, East Antarctica	143
<i>M.P. Gaur, R. Asthana and A. Chaturvedi</i>	
Mineral Chemistry, Fluid Inclusion Studies and P-T Path of Gneiss-Granulite Rocks of Schirmacher Region, East Antarctica . .	159
<i>D. Rameshwar Rao, Rajesh Sharma and N.S. Gururajan</i>	
Mapping in Antarctica	185
<i>Major C.S. Bewli</i>	

BIOLOGICAL SCIENCES

Comparative Account of Benthic Community at Two Different Locations in the Continental Antarctica	195
<i>B.S. Ingole and V.K. Dhargalkar</i>	
Ecosystem Characterization in Indian Ocean Sector, Antarctica	207
<i>S.G. Prabhu Matondkar, V.K. Dhargalkar and A.H. Parulekar</i>	
Ecobiological Studies of the Freshwater Lakes at Schirmacher Oasis, Antarctica	233
<i>X.N. Verlekar, V.K. Dhargalkar and S.G.P. Matondkar</i>	
Seasonal Variations in Carbon Budget in Water Column off Princess Astrid Coast, Antarctica	259
<i>V.K. Dhargalkar, S.G. Prabhu Matondkar and X.N. Verlekar</i>	

ENGINEERING

Engineering and Scientific Activities/Studies in Antarctica	267
<i>K.R. Sivan</i>	

MEDICAL SCIENCES

Desynchronization of Circadian Rhythms in Winterover Antarctic Expeditioners	291
<i>Nalin Mehta, Ms Naidu and U. Sachdeva</i>	

CONTENTS

xiii

Page

LOGISTICS

Report on Logistic Support	301
<i>D.V.S. Rathee</i>	
Communication Report	315
<i>H.D. Sharma</i>	
Report on IAF Operation	327
<i>A.S. Sidhu</i>	
Medical Cover	331
<i>G. Ravindranathan</i>	
The Way Thing Rolled	333

TEAM MEMBERS

S.No.	Name	Organisation
Winter Team		
1.	Dr. V.K. Dhargalkar Leader & Station Commander	National Institute of Oceanography, Goa
2.	Shri S. Venkateswarlu	India Meteorological Dept. New Delhi
3.	Shri R.P. Lal	-do-
4.	Dr. Nalin Mehta	All India Institute of Medical Sciences, New Delhi
5.	Shri K.R. Sivan	Research and Development Establishment (Engineers), Pune
6.	Shri Arun Chaturvedi	Geological Survey of India, Faridabad
7.	Shri Rajesh Asthana	-do-
8.	Lt. Col. G. Ravindranath	AMC, New Delhi
9.	Lt. Col. D.V.S. Rathee	Indian Army
10.	Lt. Col. V.S. Grewal	-do-
11.	Sub. Ajit Singh	-do-
12.	Sub. R. C. Sharma	-do-
13.	Nb. Sub. Swaran Singh	-do-
14.	Nb. Sub. T. Jambunathan	-do-
15.	Nb. Sub. Swaroop Singh	-do-
16.	Nb. Sub. K.K. Bakshi	-do-
17.	Hav. A. Raman	-do-
18.	Hav. Navinder Singh	-do-
19.	HMT K. Sadanandan	-do-
20.	HMT Vilas Shinde	-do-
21.	NK. Anand Ballabh Joshi	-do-
22.	Lt. Cdr (SDC) H.D. Sharma	Indian Navy
23.	Shri S.K. Sharma PCPOR (TEL)	-do-
24.	Shri S. Pasha POELR	-do-
25.	Shri S.P. Jothi POCK 'O'	-do-

S.No.	Name	Organisation
Summer Team		
26.	Dr. B.S. Ingole	National Institute of Oceanography, Goa
27.	Shri M.P. Gaur	Geological Survey of India, Faridabad
28.	Dr. A.N. Hanchinal	Indian Institute of Geomagnetism, Bombay
29.	Dr. Manjit Singh	National Physical Laboratory, New Delhi
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32.	Major C.S. Bewli	Survey of India, Dehra Dun
33.	Shri A. Ravichandran	-do-
34.	Shri H.K. Gaur	Defence Electronic Applications Laboratory, Dehra Dun
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49.	Sgt. R.K. Kale	-do-
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54.	Cpl. A.K. Sinha	-do-
55.	Cpl. M. Manikandhan	-do-
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